

1. Administrative & Study Identification

Full Study Title: A Phase III, Randomized, Open-Label Study Evaluating the Efficacy and Safety of Giredestrant in Combination With Phesgo Versus Phesgo After Induction Therapy With Phesgo + Taxane in Patients With Previously Untreated HER2-Positive, Estrogen Receptor-Positive Locally-Advanced or Metastatic Breast Cancer **Short Title/Acronym:** heredERA Breast Cancer **Protocol Number:** W043571 **NCT Number:** NCT05296798 **Posting ID:** 367743300117526288 **Sponsor Name:** Hoffmann-La Roche (Roche) **Trial Status:** Active, not recruiting (ACTIVE_NOT_RECRUITING) **Investigational Product:** Giredestrant (Next-generation oral SERD), Phesgo (Pertuzumab, Trastuzumab, and Hyaluronidase-zzxf), and Taxane (including paclitaxel [ATC Code: L01CD01; Trade names: Abraxane, Onxol, Taxol, Paccal] or docetaxel) **Phase of Development:** Phase III **Medical Monitor:** Clinical Operations Lead, Hoffmann-La Roche

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy of giredestrant in combination with Phesgo compared to Phesgo alone (following induction therapy with Phesgo plus taxane) measured by investigator-assessed Progression-Free Survival (PFS) in participants with locally advanced or metastatic breast cancer. **Secondary Objectives:** To evaluate Overall Survival (OS), Objective Response Rate (ORR), Duration of Response (DOR), Clinical Benefit Rate (CBR), Quality of Life (QoL), and overall safety and tolerability.

2.2 Background & Rationale Estrogen receptor (ER)-positive breast cancer accounts for approximately 70% of breast cancer cases, where tumor cells possess receptors that attach to estrogen, fueling tumor growth. Giredestrant is an investigational, oral, potent next-generation selective estrogen receptor degrader (SERD) designed to block estrogen from binding to the ER, triggering its degradation. This study is conducted to evaluate whether combining this potent targeted endocrine therapy (giredestrant) with dual HER2 blockade (Phesgo) delays disease progression and reduces treatment burden compared to standard dual HER2 blockade alone in patients with previously untreated HER2-positive, ER-positive advanced breast cancer.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional Control Type: Active Comparator (Phesgo alone) **Blinding:** Open-label **Randomization:** Randomized, Two-arm, Parallel Group

3.2 Planned Enrollment & Duration Target **N**: 922 participants (Event-driven design based on PFS) **Number of Sites:** Multicenter (Global) **Estimated Study Start:** Q2 2022 **Estimated Primary Completion:** 30-Jan-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years with an Eastern Cooperative Oncology Group (ECOG) Performance Status of 0 or 1. 2. Histologically or cytologically confirmed and documented human epidermal growth factor receptor 2 (HER2)-positive / estrogen receptor (ER)-positive adenocarcinoma of the breast with metastatic or locally-advanced disease not amenable to curative resection. 3. At least one measurable lesion and/or non-measurable disease evaluable according to Response Evaluation Criteria in Solid Tumors (RECIST) version 1.1. 4. Left ventricular ejection fraction (LVEF) $\geq 50\%$ measured by echocardiogram (ECHO) or MUGA scan. 5. Disease-free interval from completion of adjuvant or neoadjuvant systemic non-hormonal treatment to recurrence of ≥ 6 months. 6. **Maintenance Phase Entry:** Must complete a minimum of 4 to a maximum of 8 cycles of induction therapy (minimum defined as Phesgo injections + 4 docetaxel infusions, or Phesgo injections + 12 paclitaxel infusions) and achieve a minimum of stable disease.

4.2 Exclusion Criteria 1. Previous systemic non-hormonal anti-cancer therapy in the metastatic or advanced breast cancer setting (up to one line of single-agent endocrine therapy is permitted). 2. Prior treatment with a selective estrogen receptor degrader (SERD) or approved/investigative anti-HER2 agents in any treatment setting (except Phesgo/trastuzumab/pertuzumab in early settings). 3. Treatment with strong CYP3A4 inhibitors or inducers within 14 days or 5 drug-elimination half-lives prior to initiation. 4. History of malignancy within 5 years prior to screening (excluding the cancer under investigation and malignancies with negligible metastasis risk).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Initial induction therapy with Phesgo plus Taxane (requiring a minimum of 4 docetaxel infusions or 12 paclitaxel infusions). Following induction, participants are randomized to receive either: * **Arm A (Experimental):** Giredestrant + Phesgo maintenance therapy. * **Arm B (Active Comparator):** Phesgo maintenance therapy. **Route of**

Administration: Giredestrant (Oral), Phesgo (Subcutaneous injection), Taxane/paclitaxel (Intravenous). **Duration of Treatment:** Cycles continue until disease progression, unacceptable toxicity, or withdrawal of consent.

5.2 Concomitant & Prohibited Therapy Permitted: Approved LHRHa therapy (required for pre-/perimenopausal women and men for gonadal function suppression), standard supportive care for treatment-related toxicities. **Prohibited:** Strong CYP3A4 inhibitors or inducers, other investigational anti-cancer agents, and concurrent endocrine therapies outside the study protocol.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumor Imaging (RECIST v1.1), LVEF Assessment, Labs
Induction	Cycles 1+	Administration of Phesgo + Taxane (paclitaxel/docetaxel), Safety Monitoring
Randomization	Post-Induction	Randomization (1:1), First Dose of Maintenance Therapy
Treatment Cycles	Day 1 of each Cycle	Administration of assigned regimen, Safety Labs, AE review, QoL Questionnaires
Tumor Assessment	Q8W - Q12W	Efficacy Assessment (CT/MRI) per RECIST v1.1
End of Study	+30 Days post-last dose	Final Safety Follow-up, Survival Follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Progression-Free Survival (PFS), defined as the time from randomization to the first documented occurrence of disease progression or death from any cause.
Measurement Method: Investigator-assessed tumor evaluations using RECIST v1.1 criteria.

7.2 Secondary & Exploratory Endpoints * Overall Survival (OS) * Objective Response Rate (ORR) and Duration of Response (DOR) * Clinical Benefit Rate (CBR) * Patient-Reported Outcomes (Quality of Life) and Safety profile (NCI CTCAE v5.0)

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring and grading using the National Cancer Institute Common Terminology Criteria for Adverse Events (NCI CTCAE) version 5.0. **Serious Adverse Events (SAEs):** Must be reported to the Sponsor within 24 hours of investigator awareness. **Data Monitoring Committee (DMC):** An external Independent Data Monitoring Committee (iDMC) will regularly evaluate safety and efficacy signals.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) Population for all primary and secondary efficacy endpoints; Safety Evaluable Population for AE incidence. **Sample Size Justification:** Powered appropriately to detect a statistically significant improvement in median PFS between the giredestrant + Phesgo arm and the Phesgo alone arm ($\alpha = 0.05$). **Handling of Missing Data:** Standard survival analysis methodologies, including Kaplan-Meier estimations and Cox proportional-hazards models. Missing efficacy evaluations will be censored according to FDA/EMA oncology guidelines.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) guidelines and the Declaration of Helsinki. **Monitoring Plan:** Scheduled onsite and remote monitoring visits to verify source data, regulatory document completeness, and protocol compliance. **Audit Trail:** Secure, validated Electronic Data Capture (EDC) system featuring comprehensive audit trails and data clarification mechanisms.

11. Study Identifiers & Governance

Primary ID: W043571 **Registry Number:** NCT05296798 **Posting ID:** 367743300117526288 **Sponsor/Lead Organization:** Hoffmann-La Roche (Roche) **Study Title:** heredERA Breast Cancer **Official Regulatory Title:** A Phase III, Randomized, Open-Label Study Evaluating the Efficacy and Safety of Giredestrant in Combination With Phesgo Versus Phesgo After Induction Therapy With Phesgo + Taxane in Patients With Previously Untreated HER2-Positive, Estrogen Receptor-Positive Locally-Advanced or Metastatic Breast Cancer

12. Clinical Framework (Oncology Specific)

Primary Condition: Locally Advanced or Metastatic Breast Cancer (Preferred Name: Advanced Breast Cancer | MeSH Code: D001943 | SNOMED ID: 254837009) **Diagnosis Threshold:** HER2-positive, ER-positive (confirmed histologically/cytologically) **Medical Methodology:** Induction therapy followed by targeted estrogen receptor degradation + dual HER2 blockade **Optimization Target:** Delayed disease progression and minimized tumor growth (PFS)

13. Study Procedures & Methodology

Management Pathway: Standard oncological clinical pathway for advanced breast cancer **Education Protocol (HCP):** Investigator

site training on protocol compliance, RECIST v1.1 evaluation, and safety reporting **Education Protocol (Patient):** Informed consent detailing drug administration, symptom tracking, and reporting **Quality Audit Mechanism:** Independent central review of imaging and continuous clinical data monitoring

14. Observation & Follow-Up Schedule

Total Duration: Follow-up extends until disease progression, death, or study closure **Onsite Visit Cadence:** Aligned with treatment cycles (typically Day 1 of each 21-day or 28-day cycle) **Remote Monitoring Frequency:** Continuous safety and survival status follow-up post-treatment **Checklist Review Interval:** Every 8 to 12 weeks for radiologic tumor assessments

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				